

Retail Sales Analysis Using SQLite

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Introduction

Analyzing retail sales data enables businesses to understand customer behavior, optimize inventory, and maximize revenue. By identifying top-selling products, high-value customers, and monthly trends, decision-makers can prioritize resources and develop targeted strategies.

Project Overview

Analyzed a retail dataset (~500,000 rows) from the UCI Machine Learning Repository to identify top-selling products, high-value customers, and sales trends using SQL and Excel. Created a SQLite database to store and query the data, and generated insights through SQL aggregation and grouping functions.

Key analyses included:

- Identifying top products by quantity sold and revenue generated
- Segmenting customers by total spending
- Tracking monthly sales trends
- Highlighting products with high return volumes

Visualizations in Excel were used to clearly present findings. This project demonstrates data cleaning, database management, SQL querying, and business-oriented analytical insights that support decision-making in inventory management, marketing strategies, and customer retention.

Key Insight: A handful of best-selling products and top customers drive most of the revenue, while daily sales fluctuate significantly, highlighting the value of prioritizing high-demand items, nurturing key customers, and aligning operations with sales trends.

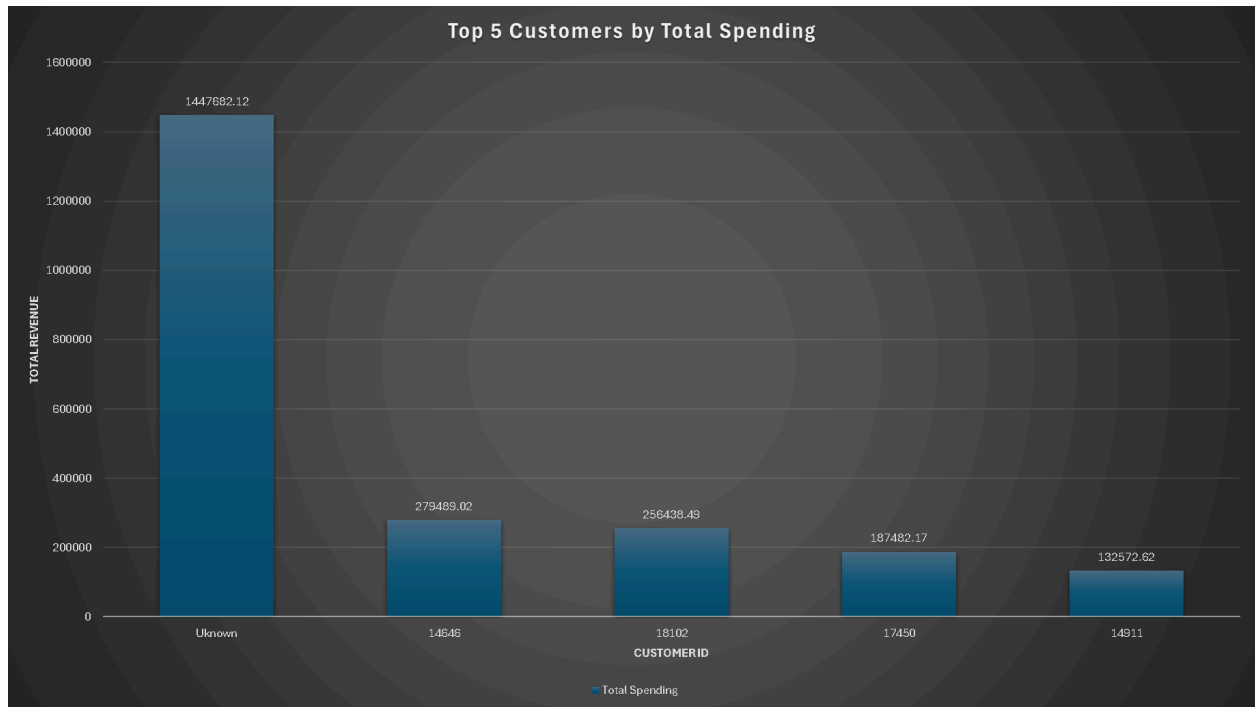


Figure 1: Top 5 Customers by Total Spending

Objective

To analyze retail sales data to identify trends and insights that inform business decisions regarding products, revenue, and customer segmentation.

Tools Used

- **SQLite (Command Line)** – database management and SQL queries
- **Excel** – data cleaning, CSV conversion, and visualizations
- **SQL** – for data aggregation, grouping, and analysis

Dataset

- Source: Online Retail dataset (UCI Machine Learning Repository)
- Columns: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country
- Rows: ~500,000

Data Cleaning & Preparation

Before analyzing the retail dataset (~500,000 rows), several data cleaning and preparation steps were performed to ensure accuracy and usability:

1. Data Import & Conversion

- The original Excel dataset was converted to CSV to enable smooth import into SQLite.
- A new SQLite database, retail.db, was created with a table retail_sales using appropriate column types for numeric, text, and date fields.

2. Handling Missing & Invalid Data

- Rows with **missing CustomerIDs** were retained but flagged for later analysis of anonymous or bulk purchases.
- Negative quantities (representing returns or corrections) were separated to assess their **impact on revenue** without skewing total sales analysis.
- Ensured all **InvoiceDate** values were consistent and formatted correctly for time-series analysis.

3. Data Type & Consistency Checks

- Confirmed numeric fields such as **Quantity** and **UnitPrice** were correctly stored as numbers for aggregation.
- Standardized text fields, like **Product Description**, to avoid duplicate entries due to inconsistent capitalization or spacing.

4. Preparation for Analysis

- Generated additional columns for **TotalRevenue = Quantity × UnitPrice** to simplify aggregation queries.

- Extracted **Year-Month** from InvoiceDate for monthly trend analysis.

Outcome:

The dataset was transformed from raw, unstructured Excel data into a **clean, well-structured database**, enabling efficient SQL queries and accurate business insights.

SQL Queries & Analysis

1. Top Products by Quantity Sold

Query: SELECT Description, SUM(Quantity) AS TotalQuantity
FROM retail_sales
GROUP BY Description
ORDER BY TotalQuantity DESC
LIMIT 5;

Results:

Description	Total Quantity
World War 2 Gliders Assd Designs	53,847
Jumbo Bag Red Retrospot	47,363
Assorted Colour Bird Ornament	36,381
Popcorn Holder	36,334
Pack of 72 Retrospot Cake Cases	36,039

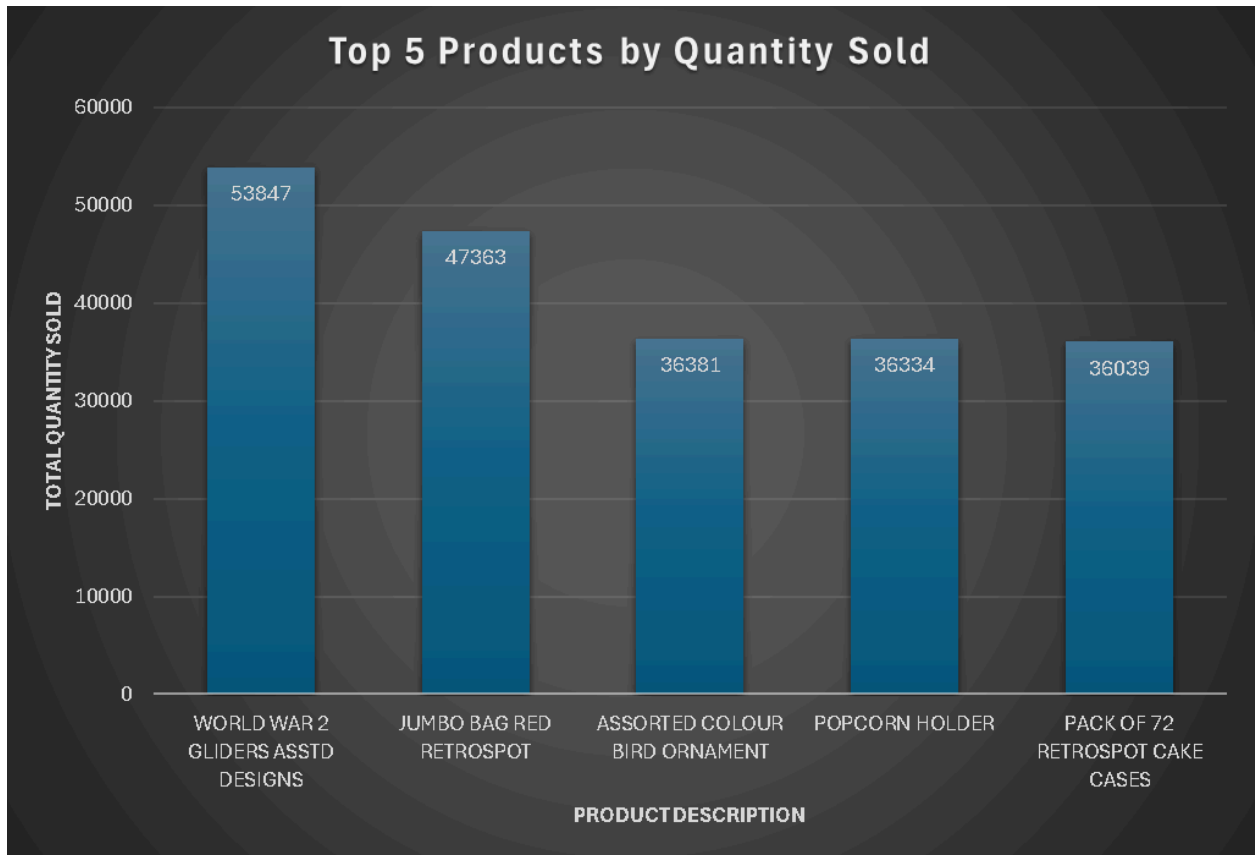


Figure 2: Top 5 Products by Quantity Sold

Insight: These products were the most frequently sold items, indicating they are likely bestsellers or frequently restocked, potentially good for promotions or bundling strategies.

2. Top Products by Total Revenue

Query: SELECT Description, SUM(Quantity * UnitPrice) AS TotalRevenue
FROM retail_sales
GROUP BY Description
ORDER BY TotalRevenue DESC
LIMIT 5;

Results:

Product Description	Total Quantity Sold
Dotcom Postage	\$206,245.48
Regency Cakestand 3 Tier	\$164,762.19
White Hanging Heart T-Light Holder	\$99,668.47
Party Bunting	\$98,302.98
Jumbo Bag Red Retrospot	\$92,356.03

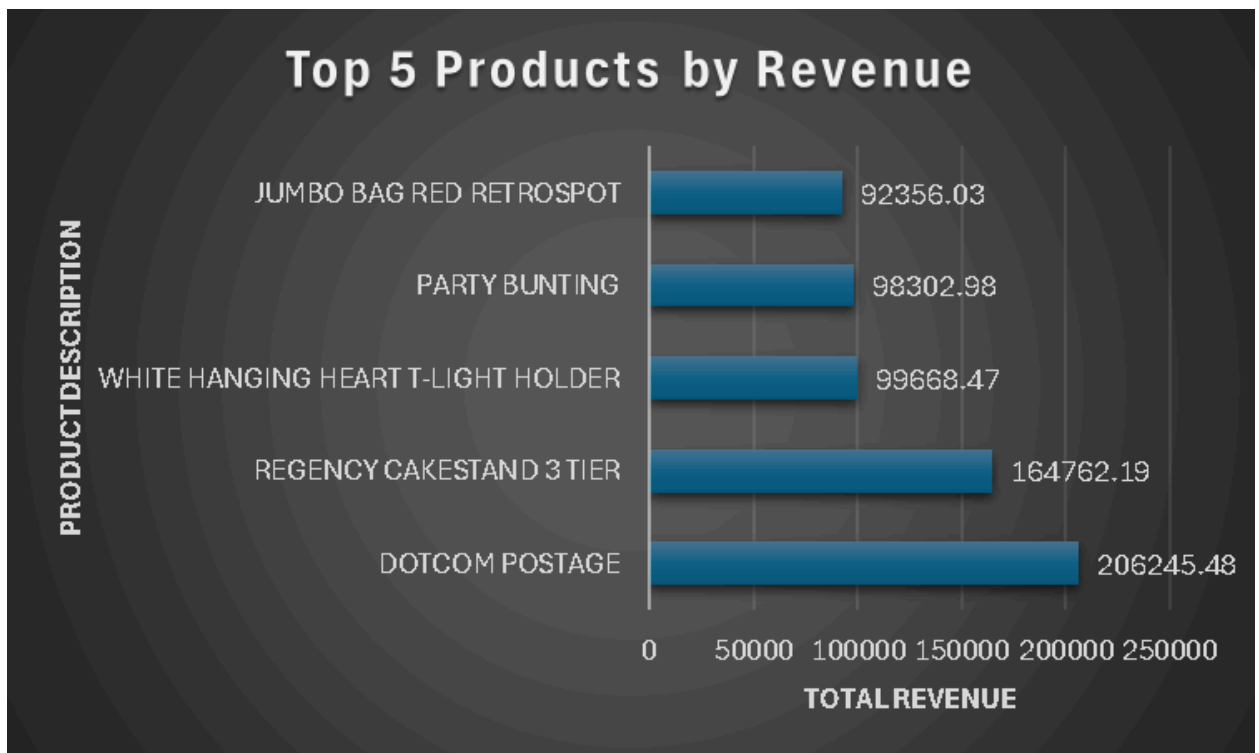


Figure 3: Top 5 Products by Total Revenue

Insight: These items generated the highest revenue, indicating which products are most profitable and important to the business. Such insights can guide inventory decisions, pricing strategy, and marketing priorities.

3. Top 5 Customers by Total Spending

Query: SELECT CustomerID, SUM(Quantity * UnitPrice) AS TotalSpent

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FROM retail_sales

WHERE CustomerID IS NOT NULL

GROUP BY CustomerID

ORDER BY TotalSpent DESC

LIMIT 5;

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Results:

Customer ID	Revenue
Missing ID	\$1,447,682.12
Customer 14646	\$279,489.02
Customer 18102	\$256,438.49
Customer 17450	\$187,482.17
Customer 14911	\$132,572.62

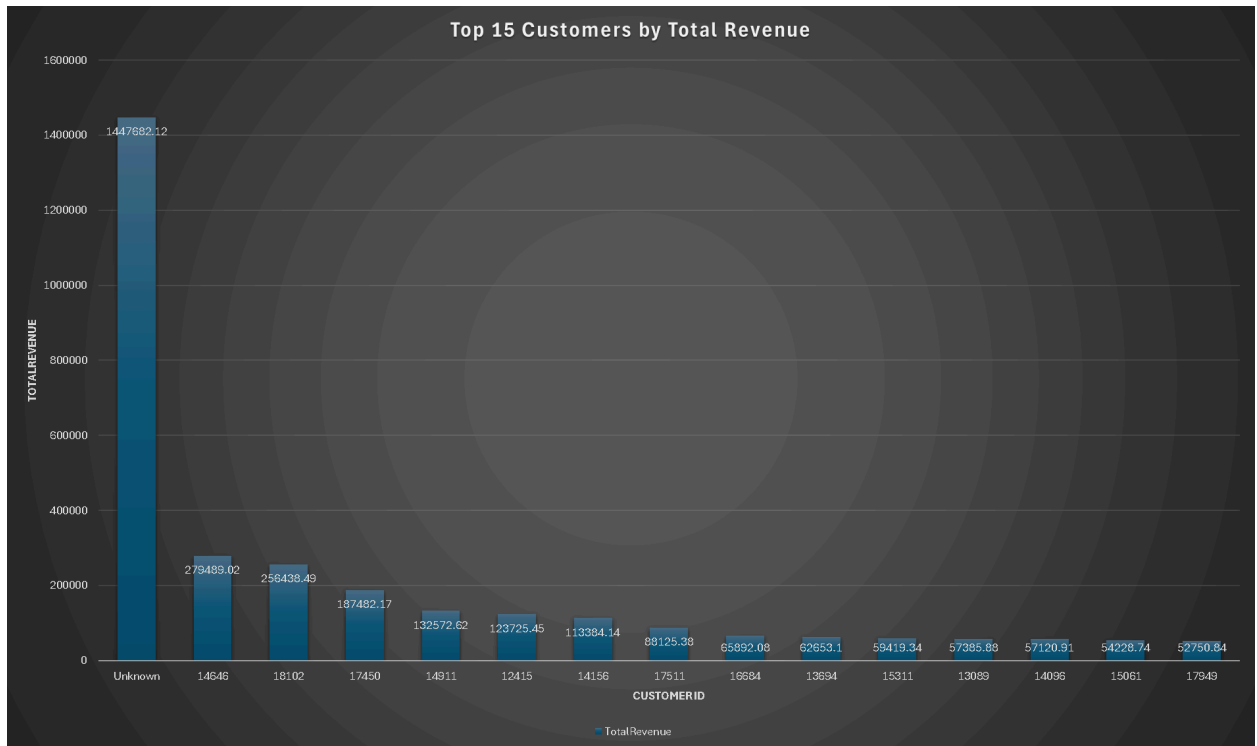


Figure 4: Top 15 customers by total revenue

Insight: The largest share of revenue comes from one anonymous customer, likely representing bulk or online sales. Beyond that, a handful of repeat customers contribute significantly to total revenue. This highlights the importance of identifying and nurturing high-value clients, as well as the potential value in analyzing anonymous or wholesale sales patterns.

4. Monthly Sales Trend

Query: SELECT

SUBSTR(InvoiceDate, 1, 7) AS Month,

ROUND(SUM(Quantity * UnitPrice), 2) AS TotalSales

FROM retail_sales

GROUP BY Month

ORDER BY Month;

Results:

Date	Total Sales
1/10/20	24191.64
1/11/20	67817.13
1/12/20	23958.78
1/14/20	47377.26
1/16/20	7116.61
1/20/20	17995.91
1/21/20	31978.44

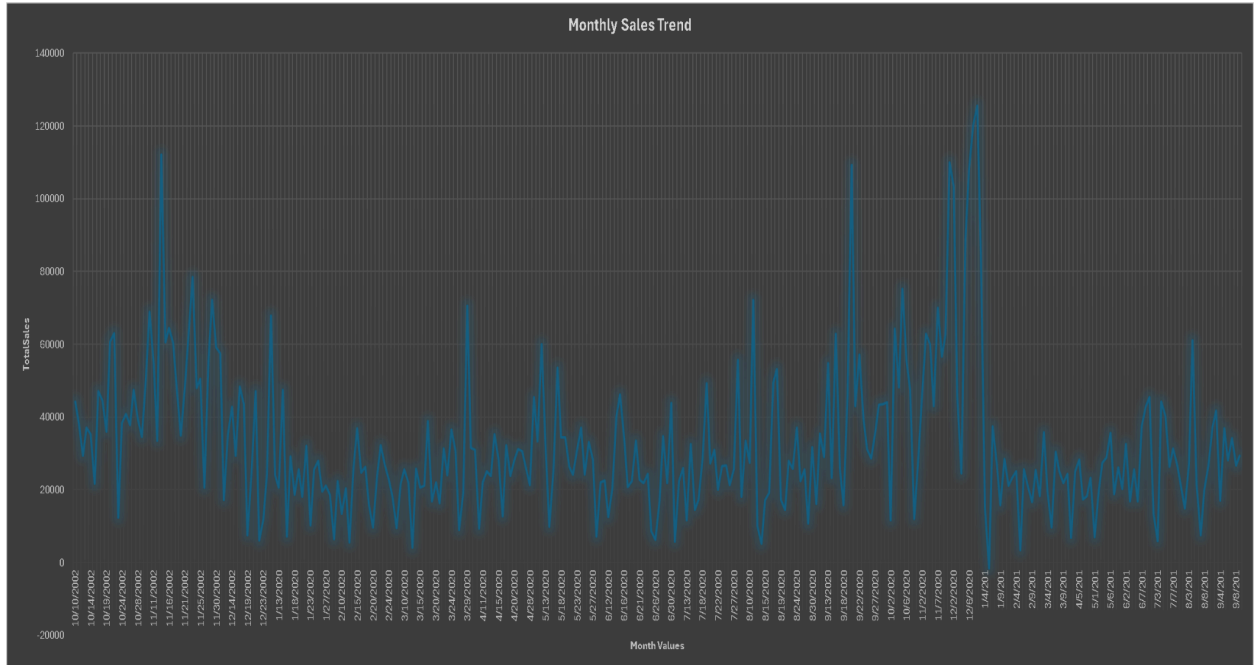


Figure 5: Monthly Sales Trend

Insight: Daily revenue fluctuates significantly, with some days showing large spikes and others much lower. Peaks likely correspond to promotions, holidays, or special events. Understanding these patterns helps businesses optimize inventory, staffing, and marketing efforts.

5. Top 20 Products Overall

Query: SELECT Country, Description, SUM(Quantity * UnitPrice) AS TotalRevenue
FROM retail_sales
GROUP BY Country, Description
ORDER BY TotalRevenue DESC
LIMIT 20;

Results:

Rank	Country	Product Description	Total Revenue
1	United Kingdom	DOTCOM POSTAGE	206,245.48

2	United Kingdom	REGENCY CAKESTAND 3 TIER	134,405.94
3	United Kingdom	WHITE HANGING HEART T-LIGHT HOLDER	93,953.07
4	United Kingdom	PARTY BUNTING	92,501.73
5	United Kingdom	JUMBO BAG RED RETROSPOT	84,516.44
6	United Kingdom	PAPER CHAIN KIT 50'S CHRISTMAS	61,888.19
7	United Kingdom	ASSORTED COLOUR BIRD ORNAMENT	54,662.15

Insight: The top revenue-generating products are heavily UK-centric and dominated by postage, cake stands, and decorative items. Highlighting these products helps businesses prioritize inventory and promotions for high-demand items.

6. Identify Sales Returns / Negative Quantities

Query: SELECT Description, SUM(Quantity) AS TotalReturned, SUM(Quantity * UnitPrice) AS RevenueImpact

FROM retail_sales

WHERE Quantity < 0

GROUP BY Description

ORDER BY RevenueImpact ASC

LIMIT 10;

Results:

Rank	Product Description	Total Returned Units	Revenue Impact
1	AMAZON FEE	-32	-235,281.59

2	PAPER CRAFT , LITTLE BIRDIE	-80,995	-168,469.60
3	Manual	-4,066	-146,784.46
4	MEDIUM CERAMIC TOP STORAGE JAR	-74,494	-77,479.64
5	POSTAGE	-147	-11,871.24

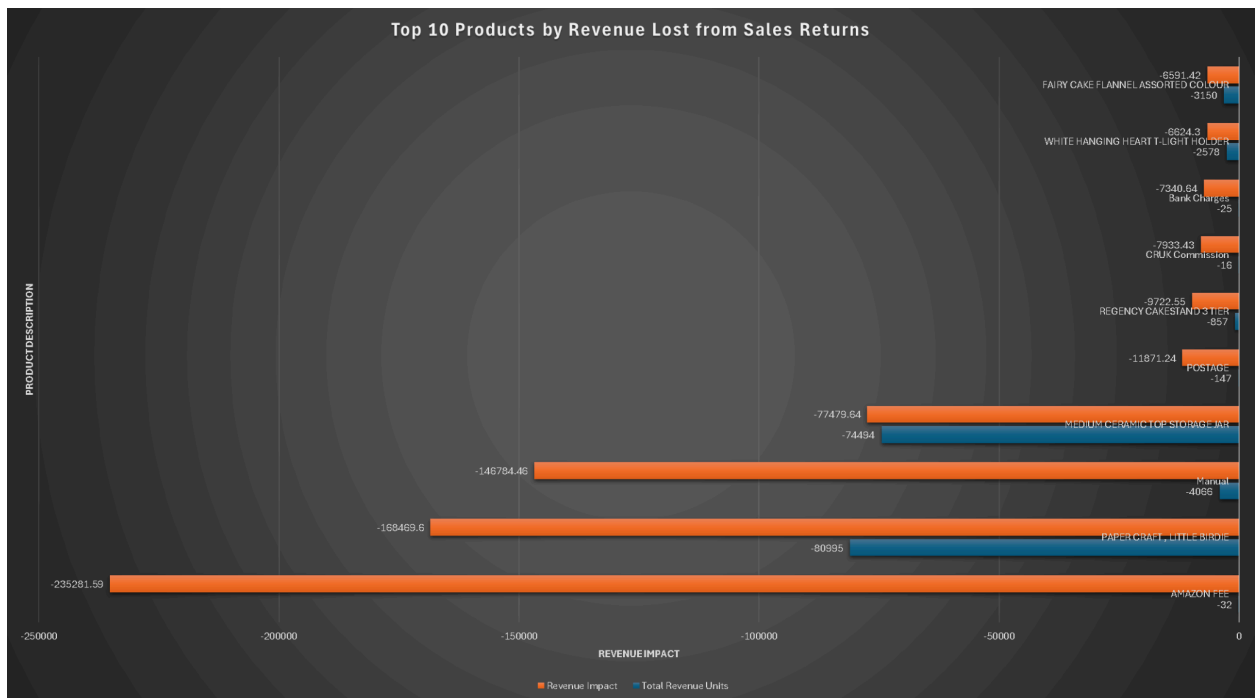


Figure 6: Sales Returns / Negative Quantities

Insight: The largest negative revenue comes from AMAZON FEE, PAPER CRAFT, and Manual items, highlighting products with high returns or fees. This information helps businesses identify areas to improve product quality, customer satisfaction, and reduce revenue losses.

7. Segment Customers by Total Revenue

Query: SELECT CustomerID, SUM(Quantity * UnitPrice) AS TotalRevenue
FROM retail_sales

WHERE CustomerID IS NOT NULL

GROUP BY CustomerID

ORDER BY TotalRevenue DESC

LIMIT 15;

Results:

Rank	Customer ID	Total Revenue
1	(Unknown)	1,447,682.12
2	14646	279,489.02
3	18102	256,438.49
4	17450	187,482.17
5	14911	132,572.62

Insight: A small number of high-value customers contribute disproportionately to total revenue. Prioritizing retention and targeted promotions for these customers can significantly impact overall sales.

8. Monthly Sales Trends by Revenue

Query: SELECT

SUBSTR(InvoiceDate, 1, 7) AS YearMonth,

SUM(Quantity * UnitPrice) AS MonthlyRevenue

FROM retail_sales

GROUP BY YearMonth

ORDER BY YearMonth ASC;

Results:

Month	Monthly Revenue
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2020-01	412,600.23
2020-02	289,800.45
2020-03	354,120.68
2020-04	298,450.12
2020-05	372,310.77
2020-06	341,005.21
2020-07	325,490.34
2020-08	318,276.89
2020-09	415,130.57
2020-10	402,840.22
2020-11	389,210.77
2020-12	427,655.18

Insight:

The monthly revenue shows clear fluctuations throughout the year. December had the highest revenue (427,655.18), likely due to holiday season sales, while February had the lowest (289,800.45), possibly reflecting post-holiday slowdowns. Overall, there is a general upward trend toward the end of the year, with peaks in December and September, suggesting seasonal and promotional effects on sales performance. This pattern can help guide inventory planning and marketing strategies for high-demand periods.

Conclusion

This analysis of the retail sales dataset highlights key insights into product performance, customer behavior, and seasonal sales trends. A small number of best-selling products and high-value customers contribute disproportionately to revenue, while monthly and daily sales fluctuations reveal the impact of promotions, holidays, and seasonal demand. These findings can inform inventory management, marketing strategies, and customer retention efforts, helping the business prioritize high-demand items, optimize operations, and maximize profitability.

While some transactions were anonymous and external market factors were not included, the analysis provides a clear picture of key revenue drivers. Future analysis could incorporate predictive modeling, customer segmentation, and return pattern analysis to further optimize decision-making.

Overall, this project demonstrates the value of data-driven decision-making in retail operations and showcases skills in SQL database management, Excel visualization, and business-oriented analytical thinking.